3. Arithmetic Progression

1. The 5th term of an A.P. is 17 and the 12th term is 38. Find the first term and common difference.

2. How many terms of the A.P. 7, 10, 13, … are needed to give a sum of 301?

3. The nth term of an A.P. is $a_n = 5 - 3n$. Find its 10th term and 25th term.

4. Find the sum of first 30 terms of the A.P. 4, 9, 14, …

5. In an A.P., the 3rd term is 8 and the 7th term is 20. Find the 12th term and the sum of first 12 terms.

6. The sum of first n terms of an A.P. is given by $S_n = 2n^2 + 3n$. Find (i) the first term (ii) the 10th term.

7. A man saves ₹200 in the first month, ₹260 in the second month, ₹320 in the third month and so on. In how many months will he save ₹9,720 in total?

8. The 9th term of an A.P. is zero. If its 5th term is 12, find its first term and common difference.

9. The sum of first 15 terms of an A.P. is 525 and the sum of the next 15 terms is 1200. Find the first term and common difference.

10. The interior angles of a polygon are in A.P. with common difference 5°. The smallest angle is 120° and the largest is 170°. How many sides has the polygon?